

Slide Courtesy : Anany Levitin

ALGORITHM *LomutoPartition*($A[l..r]$)

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//Partitions subarray by Lomuto's algorithm using first element as pivot
//Input: A subarray  $A[l..r]$  of array  $A[0..n - 1]$ , defined by its left and right
//      indices  $l$  and  $r$  ( $l \leq r$ )
//Output: Partition of  $A[l..r]$  and the new position of the pivot
 $p \leftarrow A[l]$ 
 $s \leftarrow l$ 
for  $i \leftarrow l + 1$  to  $r$  do
    if  $A[i] < p$ 
         $s \leftarrow s + 1$ ;  $\text{swap}(A[s], A[i])$ 
 $\text{swap}(A[l], A[s])$ 
return  $s$ 
```

ALGORITHM *Quickselect*($A[l..r]$, k)

//Solves the selection problem by recursive partition-based algorithm

//Input: Subarray $A[l..r]$ of array $A[0..n - 1]$ of orderable elements and

// integer k ($1 \leq k \leq r - l + 1$)

//Output: The value of the k th smallest element in $A[l..r]$

$s \leftarrow \text{LomutoPartition}(A[l..r])$ //or another partition algorithm

if $s = k - 1$ **return** $A[s]$

else if $s > l + k - 1$ *Quickselect*($A[l..s - 1]$, k)

else *Quickselect*($A[s + 1..r]$, $k - 1 - s$)

EXAMPLE Apply the partition-based algorithm to find the median of the following list of nine numbers: 4, 1, 10, 8, 7, 12, 9, 2, 15. Here, $k = \lceil 9/2 \rceil = 5$ and our task is to find the 5th smallest element in the array.

We use the above version of array partitioning, showing the pivots in bold.

0	1	2	3	4	5	6	7	8
<i>s</i>	<i>i</i>							
4	1	10	8	7	12	9	2	15
	<i>s</i>	<i>i</i>						
4	1	10	8	7	12	9	2	15
	<i>s</i>						<i>i</i>	
4	1	10	8	7	12	9	2	15
		<i>s</i>					<i>i</i>	
4	1	2	8	7	12	9	10	15
		<i>s</i>						<i>i</i>
4	1	2	8	7	12	9	10	15
2	1	4	8	7	12	9	10	15

Since $s = 2$ is smaller than $k - 1 = 4$, we proceed with the right part of the array: